Derived Units

Exam code: 9MA0

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Derived units

What are derived units?

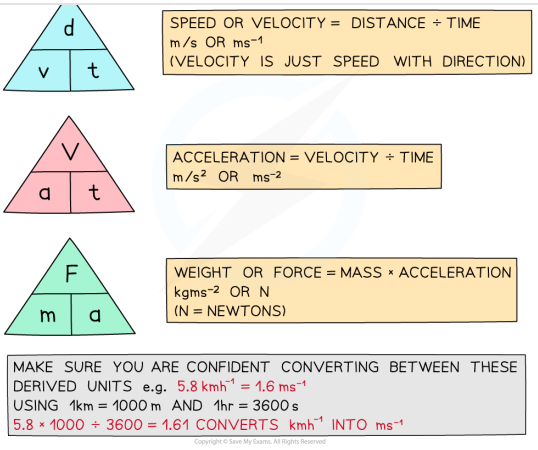
- Derived units for quantities such as velocity, acceleration and force etc. are combinations of S.I. units
  - **Speed** or **Velocity** = distance ÷ time = **m/s or m s^{-1}** (Velocity is just speed with direction)
  - **Acceleration** = velocity ÷ time = ms^{-1} /s = **m/s^{2} or m s^{-2}**
  - **Weight** or **Force** = mass x acceleration = **kg m s^{-2} or N** (N = Newtons)

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

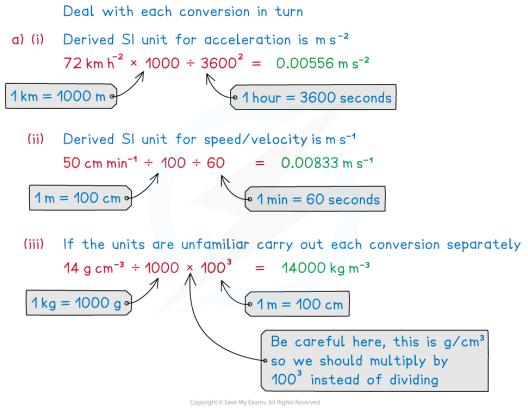


Worked Example

a) Convert the following into S.I. units

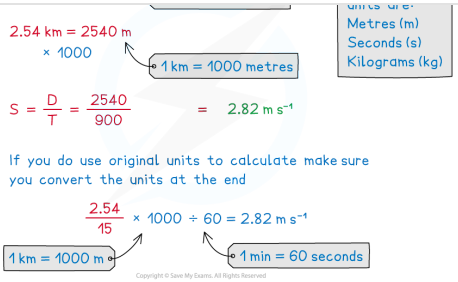
i) 72 km h^{-2} ii) 50 cm per minute iii) 14 g cm^{-3}

Answer:



b) A cyclist takes 15 minutes to travel 2.54 km. Calculate the average speed in m s^{-1}?

Answer:



Examiner Tips and Tricks

- Watch out for exam questions using non S.I. units such as feet and inches or miles per hour.
- Make sure you are confident **converting** between these derived units
- Remember **speed** is velocity without direction and **weight** is a force measured in **Newtons**, not a mass measured in kg.
- If you ever forget a formula, use the units it is measured in to figure it out
- e.g. m/s = metres ÷ seconds = distance ÷ time = velocity.

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